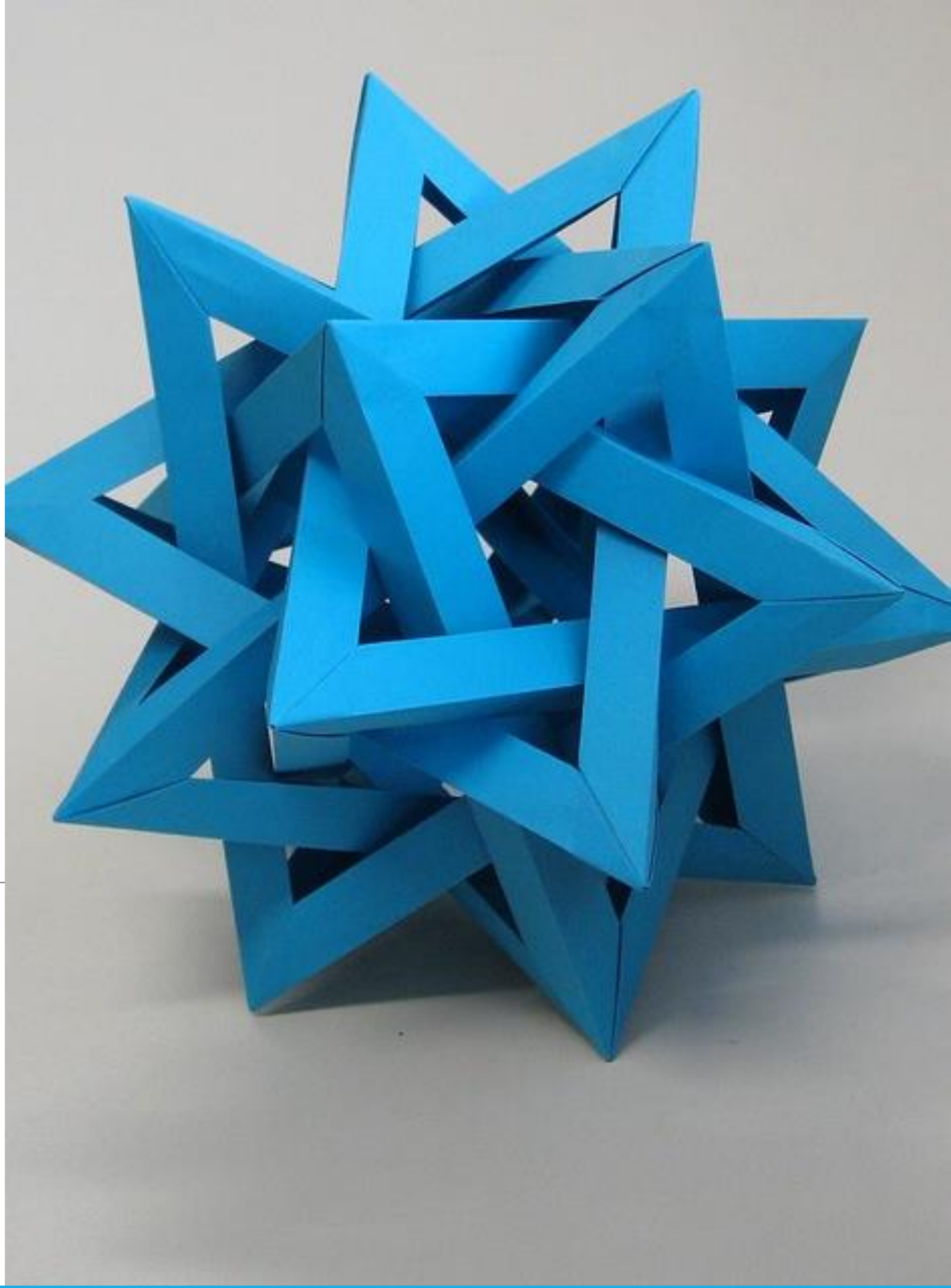
LABORATORY 10

FILES



Recommended exercises

10.1.2

10.1.4

10.2.2

DOWNLOAD THE ZIP
WITH THE DATA FILES TOO

General Guidelines

Be careful how you open files (read/write/both…)

Ask yourself the question:

- **Is it necessary to store the entire file or can the problem be solved by working on one line at a time?**

Having understood this, **what type of reading scheme should we then use?**

Remember to always close the file when you’ve finished to use it.

Exercise 10.1.2

Read request from Lab text…

Exercise 10.1.2

We can only read text files ***sequentially***

- From start to finish
- Without jumping arbitrarily either forward or backward

The only way to solve this problem is to store the entire file

- But we still need to be able to access the content line by line.
  - Which method?
- Once stored in a list we can iterate as we wish

Exercise 10.1.4

Read request from Lab text…

Exercise 10.1.4

Example **file.txt**

```
Bob;Dinner;10.00;January 1, 2013
Tom;Dinner;14.00;January 2, 2013
Anne;Lodging;125.00;January 3, 2013
Jerry;Lodging;125.00;January 4, 2013
```

```
Valid file with output:
Dinner 24.00
Lodging 250.00
```

invalid_file.txt

```
Bob;Dinner;10.00;January 1, 2013
Tom;Dinner;14.00;
Anne;Lodging;PIPPO;January 3, 2013
Jerry;Lodging;125.00;January 4, 2013
```

Invalid file:

- A field is missing in line 2
- The price in line 3 is not a float.

Exercise 10.1.4

After understanding how to process the file...

Check :

- If each row contains 4 elements
  - Beware of the 'non-standard' separator
- We will use a data structure (eg list, table?)
  - **For service types**
  - **For the totals**

Each line in the file will correspond to one of two cases:

- **An existing service** -> We only need to update the total cost
- **A service that is not there** -> We need to add the service and update the cost

Exercise 10.2.2

Read request from Lab text…

Exercise 10.2.2

Solution structure

Read ID

For each line of the classes.txt file (which corresponds to a file name) :

***Open it** (pay attention to generate the full name with the ".txt"!)*

Scan it:

if the ID number is contained in the file, print the grade

Print in the requested format!!!

You can complicate the exercise a little by generating an output file where you save this information.